

claude-windows / c6028e72-ebfd-4218-89a8-c4d7dd23be96

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\c6028e72-ebfd-4218-89a8-c4d7dd23be96\subagents\workflows\wf_f9aa1d1c-c83\agent-  
a580322e9dec49d24.jsonl`  
- SHA-256: `d71b0a6724a2cb6befeb19146a8485048bfc56dda0ee11f01dcb7472ef82ddd9`  
- Source modified: `2026-06-09T11:29:48+00:00`  
- Imported at: `2026-07-05T16:48:23+00:00`  
- project: `wf_f9aa1d1c-c83`  
- session_id: `c6028e72-ebfd-4218-89a8-c4d7dd23be96`
```

Transcript

1. user (2026-06-09T11:25:35.355Z)

DIMENSION = MODEL STRATEGY for HIGH precision/recall rally+highlight EXTRACTION from a clip, MULTISPORT, single model. Use WebSearch+WebFetch. Can a fine-tuned VLM reach HIGH P/R on temporal localization (vs specialized temporal-action-localization heads like ActionFormer/ASFormer)? Trade-offs: a tiny per-task TAL head is cheap + high-P/R but can't do conversation/multisport-generalization; a VLM does conversation + generalizes but is harder to get to high P/R + costly. Recommend: start with TAL head, VLM, or BOTH (e.g., TAL/own-model as a label-amplifier/teacher + baseline, VLM as the target). Data/compute for each; multisport single-model feasibility (one model for badminton/tennis/TT/pickleball/padel).

OWNER CONTEXT:

OWNER'S REFRAMED VISION (from PR #61 review comments ? supersedes earlier framing where they differ):

- IMMEDIATE GOAL: a TRAINING FRAMEWORK + HARNESS to produce a VERY HIGH precision/recall model that extracts

- HIGHLIGHTS and RALLIES from a clip. Quality first. On-device is LATER (a model-compression goal for power users to

- save upload cost) ? NOT the immediate goal; don't over-index on it now.

- MULTISPORT FROM THE START: all NET-SEPARATED RACQUET SPORTS (badminton, tennis, table tennis, pickleball, padel)

- first, eventually ALL sports. A SINGLE, sports-AGNOSTIC model (the collector + indexer are already sport-generic by

- design; the repo will be renamed off "badminton"). More labels/compute/storage is an accepted "good problem."

- CONVERSATIONAL VIDEO-QA is a north-star feature: upload a video -> the model processes it -> the user asks CONTEXTUAL

- QUESTIONS continuously, e.g. "show me the longest rally", "make a video of all the service faults in this clip",

- "show me a clip where the rally lasted over 20 shots". This implies per-video

- STATE/bookkeeping the model itself

- won't hold ? so the FRAMEWORK must be designed for that bookkeeping from the start. Owner asks: feasible to embark on

- now, or punt and revisit after quality is solid?

- HARD COMMERCIAL-LICENSE GUARDRAIL (owner: "never corrupted"): EVERY dependency ? code, data, AND model weights ? must

- be usable in PRIVATE PROPRIETARY COMMERCIAL software. MIT / Apache-2.0 / BSD are the easy fits. Reject viral/NC/

- custom-restrictive (e.g. Gemma-3 viral terms; weights trained on paid-LLM outputs; >X-MAU caps if truly binding).

- NEVER train on paid-LLM (Gemini/OpenAI/Anthropic) outputs. Exclude minors; separate training opt-in. Hopefully a

- CONVERSATIONAL single model.

- CURRENT STATE (de-risking evidence): on a 6-match human-labeled corpus, a learned per-frame prototype gets held-out

- AUC 0.83-0.92 and a cross-video LOVO mean 0.583 vs the heuristic's 0.480 ? and the learned

model DECOUPLES from the

multi-court "background games" confound the heuristic can't escape. WASB = the shuttle detector (MIT code, but the

distributed checkpoint is academic-data-trained ? provenance unresolved = a commercial-cleanliness blocker to fix).

- CONSTRAINTS: solo founder, Bangalore; local dev box = ONE RTX 2070 Super (8GB) + WSL2; paid Gemini stays the fallback;

cloud GPU rented per-job (Modal/RunPod per the earlier study).

REQUIREMENTS BRIEF:

```
{"requirements":["GOAL (reframed): The immediate deliverable is a TRAINING FRAMEWORK + HARNESS that produces a VERY high precision/recall model for extracting HIGHLIGHTS and RALLIES from a clip. Quality is the first-order objective; everything else is secondary to model quality.", "ON-DEVICE IS DEFERRED (reframe vs earlier 'on-device = immediate moat'): On-device / model-compression is a LATER goal, motivated as an upload-cost saver for power users. It is NOT the immediate target and the strategy must NOT over-index on it now. (Supersedes OWNED_MODEL_IMPLEMENTATION_PLAN.md's 'on-device = the privacy moat / immediate' framing.)", "MULTISPORT FROM DAY ONE: Build for ALL net-separated racquet sports first (badminton, tennis, table tennis, pickleball, padel), with a path to eventually ALL sports. No single-sport (badminton-only) scoping.", "SINGLE SPORT-AGNOSTIC MODEL: Target ONE sports-agnostic model rather than per-sport models. Collector + indexer are already sport-generic by design; the repo will be RENAMED off 'badminton'. Generic data schema stays; coach->athlete remains an optional overlay, not baked in.", "SCALE COST ACCEPTED: More labels / compute / storage from multisport breadth is an explicitly accepted 'good problem' ? do not constrain the design to minimize label/compute/storage volume at the expense of breadth or quality.", "CONVERSATIONAL VIDEO-QA IS THE NORTH-STAR FEATURE: upload a video -> model processes it -> user asks CONTEXTUAL questions CONTINUOUSLY (e.g. 'show me the longest rally', 'make a video of all the service faults', 'show me a clip where the rally lasted over 20 shots'). Hopefully served by a single conversational model.", "PER-VIDEO STATE/BOOKKEEPING IS A FRAMEWORK REQUIREMENT: The conversational-QA loop implies durable per-video state (rally inventory, shot counts, event/fault tags, timestamps, derived clips) that the MODEL ITSELF will not hold. The FRAMEWORK must be DESIGNED FOR THIS BOOKKEEPING FROM THE START, even if the QA feature ships later.", "OPEN DECISION the owner is asking for a recommendation on: is conversational-QA feasible to EMBARK ON NOW, or should it be PUNTED and revisited AFTER rally/highlight quality is solid? (The bookkeeping-from-the-start requirement holds either way.)", "HARD COMMERCIAL-LICENSE GUARDRAIL ('never corrupted'): EVERY dependency ? code, training data, AND model weights ? must be usable in PRIVATE, PROPRIETARY, COMMERCIAL software. MIT / Apache-2.0 / BSD are the accepted easy fits.", "REJECT viral / non-commercial / custom-restrictive licenses: explicitly reject Gemma-3 viral terms, weights trained on paid-LLM outputs, and >X-MAU caps IF truly binding for commercial use.", "NEVER TRAIN ON PAID-LLM OUTPUTS: Gemini / OpenAI / Anthropic outputs must never be a training-data source (terms/copyright). Paid Gemini remains inference/serving/FALLBACK only. (Also: the existing Gemini-silver training labels in training.label_candidates must be purged before they bake in ? M0.3.)", "DATA-ELIGIBILITY GUARDRAILS: Exclude minors from the training pool; per-user training data requires a SEPARATE explicit opt-in. Split data by match.", "QUALITY EVIDENCE / STARTING POINT (de-risking baseline to beat): On a 6-match human-labeled corpus, the learned per-frame prototype achieves held-out AUC 0.83-0.92 and cross-video LOVO mean 0.583 vs the heuristic's 0.480. The learned model is the chosen path BECAUSE it DECOUPLES from the multi-court 'background games' confound that the heuristic cannot escape (3 heuristic levers ? spatial gate, audio-foreground, court-region separability ? were each tested and failed this session; the problem is temporal, not spatial).", "WASB PROVENANCE IS AN OPEN COMMERCIAL-CLEANLINESS BLOCKER: WASB is the shuttle/ball detector (MIT code), but the distributed checkpoint is academic-data-trained with unresolved provenance (the C3 item) ? this is a commercial-cleanliness blocker that must be resolved (document as Gen-0 prerequisite + plan own-weights, or scope an alternative.)", "RESOURCE/OPERATING CONSTRAINTS: Solo founder, Bangalore. Local dev box = ONE RTX 2070 Super (8GB VRAM) + WSL2 (no full-pipeline run in the dev container ? no GPU/torch/cv2/numpy there). Cloud GPU rented per-job (Modal / RunPod per the earlier study). Paid Gemini stays the fallback.", "LANGUAGE/SCOPE NOTE: The reframed vision adds no new natural-language/multilingual requirement beyond English conversational-QA phrasing; 'language' as a strategic axis is limited to the conversational-QA query interface, not localization.", "tensions":["QUALITY-FIRST vs ON-DEVICE-LATER vs HARDWARE: The very-high-quality goal pushes toward larger VLM bases and cloud training, but the only local box is an 8GB RTX 2070 Super and on-device is explicitly deferred ? so the immediate model is NOT constrained to fit 8GB, yet the eventual compression goal must remain reachable. The strategy must pick a base that is both high-quality-trainable AND later-compressible without re-architecting.", "MULTISPORT-FROM-START vs CURRENT EVIDENCE DEPTH: The de-
```

risking corpus is badminton-only (WASB shuttle features, multi-court badminton confound). A single sport-agnostic model is required from day one, but the only validated signal/features and the only labeled corpus are badminton ? the feature stack (WASB trajectory) may not generalize to tennis/table-tennis/padel, and per-sport corpora don't exist yet.", "SINGLE SPORT-AGNOSTIC MODEL vs COMMERCIAL-CLEAN DETECTORS: A sport-agnostic model likely needs per-sport object/ball detectors or a unified detector; WASB (badminton) already has unresolved provenance (C3), and clean MIT/Apache/BSD detectors for tennis/TT/pickleball/padel are not yet identified ? each new sport reopens the license-cleanliness guardrail.", "CONVERSATIONAL-QA NORTH-STAR vs 'NEVER TRAIN ON PAID-LLM': A single conversational model that answers contextual video questions typically wants an LLM/VLM teacher, but the strongest available teachers (Gemini/OpenAI/Anthropic) are forbidden as a training source ? so the conversational capability must come from an Apache-2.0/MIT base + own data, narrowing the candidate set and raising the quality bar to clear without paid-LLM distillation.", "BOOKKEEPING-FROM-THE-START vs SEQUENCING DISCIPLINE: The owner wants the framework designed for per-video QA state NOW, but the committed next slice is the async job model (single-tenant) and heavy model work is gated on scaffolding + sign-off ? designing durable QA-state bookkeeping prematurely risks building against an architecture (and a model) that doesn't exist yet (the 'embark now vs punt' question the owner is explicitly asking).", "ACCEPTED SCALE COST vs SOLO-FOUNDER / SINGLE-GPU BUDGET: 'More labels/compute/storage is a good problem' collides with a solo founder on one 8GB GPU renting cloud per-job ? multisport label volume and storage are accepted in principle but still bounded by one person's labeling throughput and per-job cloud spend.", "SINGLE CONVERSATIONAL MODEL' (hoped) vs HARNESS-FIRST QUALITY GOAL: The owner hopes for one model that does detection + segmentation + conversational QA, but the harness/quality goal is concretely about a rally/highlight extractor; folding conversational QA into the same model vs keeping extraction (model) and QA (bookkeeping + retrieval/LLM layer) separate is an unresolved architectural fork.", "QUALITY-FIRST ('very high precision/recall') vs HONEST EVIDENCE CEILING: Current best cross-video LOVO is ~0.583 (directional, $n \leq 6$, linear, threshold-transfer unsolved) ? a large gap from 'very high precision/recall', so the quality target may require substantially more data/features/base-model than the current de-risking artifacts demonstrate."}]}

2. assistant (2026-06-09T11:25:39.875Z)

I'll research this thoroughly. Let me start by reviewing the relevant project docs to ground my recommendation in the current state, then fan out web searches on the VLM-vs-TAL temporal localization question.

3. user (2026-06-09T11:25:41.290Z)

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md

4. user (2026-06-09T11:25:41.310Z)

docs\EVALUATION.md
docs\TRACKNET_WSL_SETUP.md
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\README.md
docs\DEFERRED_HARDENING_PLAN.md
docs\PMF_MARKET_RESEARCH.md
docs\ROADMAP.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\decisions\DECISIONS.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\past_projects\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\COMMERCIALIZATION.md

docs\PLATFORM_ARCHITECTURE.md
docs\BACKLOG.md
docs\GOLDEN_SET_VENDORING.md
docs\README.md
docs\VIDEO_RECORDING_GUIDELINES.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\CODE_MAP.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\quality-iterations\README.md

5. assistant (2026-06-09T11:25:44.696Z)

Let me read the two owned-model docs and the current-state memory to ground my recommendation, then fan out web research.

6. user (2026-06-09T11:25:45.670Z)

```
1      # Owned-model implementation plan (DRAFT ? for owner review, no code started)
2
3      **Status:** PLAN ONLY. No owned-model implementation begins until the owner greenlights
a specific
4      milestone (per CLAUDE.md). Paid Gemini stays the always-ready primary/fallback
throughout.
5      **Grounded in:** `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` (the verified
study) +
6      `docs/archives/quality-iterations/2026-06-multivideo-lovo/` (the LOVO + prototype that
de-risk it).
7
8      ## North star & non-negotiables
9      - **Goal:** an OWNED, on-device rally?highlight model; local inference = the privacy
moat. The moat is the
10     consented dataset + eval harness, **not** the weights.
11     - **Guardrails (hard):** Apache-2.0/MIT bases only (Gemma-3 rejected); **never** train
on paid-LLM
12     (Gemini/OpenAI/Anthropic) outputs; exclude minors + separate training opt-in; split
data **by match**.
13     - **Sequencing discipline:** committed next platform slice is the **async job model
(scaffolding)**. Heavy
14     model *training* (Gen-2, cloud) waits for P0?P4 scaffolding + a healthy corpus +
explicit go. The
15     **corpus + data + compliance foundation is no-regret and can proceed first** ? it's
what makes later
16     training cheap and clean.
17
18     ## Why now is the right moment (today's evidence)
19     - Heuristic cross-video LOVO =  $0.443 \pm 0.279$  (Adarsh 0.722, Kushagra 0.163-ceiling).
The  $\pm 0.279$  is
20     *method* variance ? the unsupervised windowing is footage-fragile even when WASB
recall is excellent
21     (Kushagra: 97% of rallies have shuttle motion, yet unsegmentable).
22     - The learned prototype gets **held-out per-frame AUC 0.84?0.88** ; Kushagra oracle F1
**0.578 vs 0.163**.
23     ? The rally signal is *learnable*; the bottleneck is **data scale + calibration (n?5),
not method**.
24
25     ---
26
27     ## Phase 0 ? Foundation (start now; no GPU/scaffolding dependency)
28
29     ### M0.1 ? The labeled-corpus spine (the moat) . *owner-greenlight item*
30     - **What:** one canonical JSONL row per rally: `{video_id (file MD5), rally_ordinal,
```

```

sport, start, end,
31     ending_reason, label_type, source(human|pseudo|vendor), annotator/provenance,
consent/training_opt_in,
32     split(train|val|test, assigned BY MATCH), trajectory_ref, labeled_window}`.
33     - **Where:** authored in `sports-data-collector` (system-of-record; extends PR #13 + the
positioning doc);
34     the indexer only consumes transcoder outputs.
35     - **3 deterministic transcoders (one row ? three views, zero re-labeling):** (1)
heuristic/LogReg
36     candidate labels (already exists: `training.label_candidates`), (2) TAL ActivityNet-
JSON, (3) VLM clip?QA.
37     - **Guardrail enforcement:** provenance/consent/minors fields gate corpus membership
*before* transcoding;
38     Gemini labels never enter a training transcoder; the new `eval_partial_labels` window
flows into every view.
39     - **One scorer spine:** `metrics.match_intervals` for heuristic-labeling, TAL eval, and
the Gen-2 reward.
40     - **Acceptance:** round-trip test (collector export ? each transcoder ? indexer
loaders); split-by-match
41     enforced; no Gemini-sourced rows in any training view.
42
43     ### M0.2 ? Grow n?2 ? 5 distinct matches . *owner action + my scoring loop*
44     - **What:** tag ?3 more *distinct* matches (gBaaddy, TestLarge, + a small Rooru batch).
I generate the
45     trajectory + score each as it lands (the loop we've been running).
46     - **"Healthy corpus" bar:** ?5 distinct match-sessions with full human labels ? stable
leave-one-match-out.
47     - **Acceptance:** n?5; per-match + LOVO numbers recorded; corpus diversity
(camera/court/lighting) noted.
48
49     ### M0.3 ? Compliance cleanup (do before it bakes in) . *owner-greenlight item*
50     - **(i) Purge Gemini-derived TRAINING labels.** The distilled pipeline currently trains
on Gemini silver
51     (`training.label_candidates` over Gemini CSVs) ? this **violates the no-paid-LLM-
training rule.** Retarget
52     training to human + open-teacher (WASB/own-model) labels only; Gemini stays
eval/teacher-reference.
53     - **(ii) WASB checkpoint provenance (C3).** The distributed
`wasb_badminton_best.pth.tar` is academic-data-
54     trained (provenance unresolved) ? it propagates to anything trained on its features.
**Decide:** document
55     as a hard Gen-0 prerequisite and plan ROADMAP Phase-4 "train own WASB weights," or
scope an alternative.
56     - **Acceptance:** no paid-LLM outputs in any training path; a written C3 decision +
prerequisite flag.
57
58     ### M0.4 ? Gen-0 A/B harness (built here; trains on your GPU box) . *owner-greenlight
item*
59     - **What:** productionize the prototype into a real harness ? feature stack (visibility-
run-length, velocity,
60     acceleration, net-crossing flags, smoothing/windowing) ? **ActionFormer (MIT, 1:1
rally=instance)** and/or
61     **ASFormer (MIT, TAS)** train?eval?ship-gate**. Reuse `metrics.match_intervals`.
62     - **Ship gate (the honest bar):** held-out per-match F1 must beat the **honest cross-
video heuristic LOVO
63     (~0.44)**, not the fit-on-self 0.73. nanochat-style pass/fail verdict.
64     - **Where it runs:** training/eval on the **GPU/WSL machine** (can't run in this dev
container). Harness +
65     feature stack are authored here; you run the A/B.
66     - **Acceptance:** one-command train?eval?verdict; reproducible; the prototype's AUC
translated to segment F1
67     at n?5; documented A/B (ActionFormer vs ASFormer vs heuristic).
68
69     ---
70

```

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71     ## Phase 1 ? Gen-0 ship decision (gated on n?5 + M0.4)
72     - Run the A/B; if a learned Gen-0 clears the honest LOVO bar with per-video calibration,
**it becomes the
73     rally segmenter** (still fully local, MIT, on the 2070 Super). If not, the finding
(more data / better
74     features needed) is itself the next-step signal ? no over-fitting to declare victory.
75     - **Decision gate:** Gen-0 adopted ? more data/features first. Either way, the corpus +
harness persist.
76
77     ## Phase 2 ? Gen-1 / Gen-2 (gated on platform scaffolding P0?P4 + explicit go)
78     - **Gen-1 (Option A):** concatenate fully-local cue channels (audio-onset, pose/court)
onto the same head;
79     fusion in our rally layer. License/cost-vet the pose extractor (open item).
80     - **Gen-2 (Option B):** e2e multimodal VLM on an Apache-2.0 base (**Qwen3-VL-4B**
primary / Qwen2.5-VL-7B
81     fallback / **Gemma 4** end-state once a timestamp recipe is proven), SFT cold-start ?
GRPO/RFT with a
82     temporal-IoU-vs-golden reward. **Cloud training on Modal** (best no-access contract);
base license
83     re-verified per-checkpoint.
84     - **Hard pre-serving gate:** the **$/call crossover benchmark** (self-hosted-VLM vs
Gemini on real workload)
85     + an **on-device latency/throughput** measurement of the quantized base on the 2070
Super ? both currently
86     *unmeasured* and load-bearing. Don't stand up an owned serving stack until these clear
(COMMERCIALIZATION C7).
87
88     ---
89
90     ## Decisions I need from you (to unblock specific milestones)
91     1. **Greenlight which Phase-0 milestone(s) to start in code** (M0.1 corpus spine = my
recommendation; M0.3
92     cleanup; M0.4 harness). M0.2 (tagging) runs regardless.
93     2. **WASB C3 stance** ? accept "document as prerequisite" for now, or prioritize own-
WASB-weights sooner?
94     3. **Ratify "Apache-2.0/MIT"** (the study's approved list includes MIT bases/teachers;
the guardrail text
95     says "Apache-2.0 only").
96     4. **Corpus home** ? confirm the collector is the system-of-record for the spine
(consistent with PR #13).
97
98     ## Open prerequisites (carried from the study's critic ? none block Phase 0)
99     WASB C3 provenance · Gemma-4 license stays AMBER until a raw HF LICENSE check · on-
device VLM latency
100    unbenchmarked · real VLM VRAM/token cost on our 1080p/4K-60fps clips · DPA + no-train
for any cloud training ·
101    the owned-vs-Gemini $/call crossover volume.
102
103    ---
104    *This plan is design-only and reversible. Nothing here starts until you greenlight a
milestone.*
105

```

7. user (2026-06-09T11:25:46.110Z)

```

1     # Owned-model training study (rally segmentation ? end-to-end multimodal)
2
3     **Date:** 2026-06-08 · **STATUS: DESIGN BLUEPRINT (no implementation until PLATFORM
P0?P4 + healthy corpus; owner sign-off required).**
4
5     > How to train our **OWNED** rally/highlight model **reusing the videos we label**,
aligned to the
6     > Apache-2.0 / no-paid-LLM-training / privacy-local guardrails. Produced by an 11-agent
research
7     > workflow (ground ? 5 web-research dimensions ? adversarial verification of

```

costs/licenses/privacy ?

8 > synthesis + completeness critic; 35 sources). Empirical results from THIS session (LOVO + a learned

9 > prototype) are folded in and ****resolve the study's biggest open unknown**** ? see \$"What we proved today".

10 > Numbers from web sources are 2026 snapshots in a $\pm 15\% \pm 25\%$ band; ****re-quote live before budgeting.****

11

12 ## Thesis

13

14 Train the owned model in ****three staged generations off ONE model-agnostic labeled corpus**** ? a label

15 collected once feeds every generation, so labels are never re-collected. ****The moat is the consented**

16 dataset + the eval harness, never the weights.**

17

18 | Gen | When | Approach | Base / model | Compute |

19 |---|---|---|---|---|

20 | ****Gen-0**** | now ? unblocks at n?5 matches | tiny ****temporal action segmentation (TAS)**** head over a hand-built per-frame feature stack from the WASB trajectory we already produce; per-frame rally(1)/non-rally(0) ? merge into [start,end] | ****ASFormer (MIT, ~1.1M params)**** primary; ****ActionFormer (MIT)**** A/B challenger (overtakes once corpus = hundreds of matches) | ****local RTX 2070 Super****, trains in minutes, \$0 |

21 | ****Gen-1 / Option A**** | 6?12 mo | concatenate more local cue channels onto the SAME head: trajectory + audio-onset + pose/court serve-gating; fusion in our rally layer, cues swappable | same ASFormer/ActionFormer, wider input | local + occasional cloud burst (pseudo-labeling) |

22 | ****Gen-2 / Option B**** | 12?24+ mo, after scaffolding + healthy corpus | end-to-end multimodal VLM: (frames+audio+pose+trajectory) ? rally [start,end] (+semantics); SFT cold-start then ****GRPO/RFT** with a temporal-IoU-vs-golden reward** | ****Qwen3-VL-4B-Instruct**** (Apache-2.0, on-device) primary; ****Qwen2.5-VL-7B**** fallback; ****Gemma 4 E4B/12B-Unified**** (Apache-2.0 + native video+audio) end-state | rented 24?48 GB A100/L40S (does NOT fit 8 GB box) |

23

24 ## What we proved today (de-risks the blueprint)

25

26 The study's ****"dominant empirical unknown"***** was: ****can a low-dim WASB trajectory (?3?4 raw dims) feed a**

27 TAS head, given ASFormer's small-data result used 2048-d I3D features?* This session answered it.

28

29 - ****Cross-video LOVO of the current heuristic**** (`calibrate\_local`, Adarsh-96 + Kushagra-32, human labels):

30 mean held-out F1 ****0.443 $\pm$ 0.279****; Adarsh 0.722, ****Kushagra 0.163**** (its own ceiling ? ****no* velocity-windowing**

31 config does better). ****The ± 0.279 spread is method variance, not config.**** ? This also ****corrects the study's**

32 ****~0.73 bar*****: that was a fit-on-self single-match number; the honest cross-video heuristic bar is ****~0.44****,

33 which ****lowers* the Gen-0 ship gate.**

34 - ****A learned per-frame prototype**** (`backend/eval/rally\_seq\_proto.py`: 6 hand-built features ? visibility rate,

35 mean/max shuttle speed, ****net-crossing rate****, x/y spread ? into the repo's dependency-free LogisticRegression,

36 honest LOVO): ****per-frame AUC 0.878 (Adarsh) / 0.841 (Kushagra), held-out.**** On Kushagra ? the match the

37 heuristic ****cannot* segment (0.163) ? a calibrated threshold reaches **segment F1 0.578 (3.5x)****. The naive

38 cross-video threshold gives only 0.049, so ****the remaining gap is threshold/operating-point calibration at n=2,**

39 ****not representation.****

40

41 - ****n=3 update (doubles match added ? `LargeTestVideo`, 49 rallies):**** the learned prototype ****overtakes the**

42 heuristic.** Mean held-out LOVO F1 ? learned ****0.377 ? 0.615****, heuristic ****0.443 ? 0.544****. Kushagra's learned

43 held-out F1 jumped ****0.049 ? 0.417**** (one added match closed most of the calibration gap); the ****doubles**** match,

44 held out and trained only on the two singles, scored AUC ****0.914**** / F1 0.729 (generalizes across play type +

45 frame rate). The heuristic still cannot rescue Kushagra at any n (0.151, capped). ****The honest "bar to beat" is**

46 thus ~0.54 (n=3), not 0.73 ? and the learned path already clears it on the mean and keeps rising with n.**

47

48 ****Conclusion:**** the rally signal IS learnable from the WASB trajectory features, cross-footage (AUC ~0.85?0.91) ?

49 the heuristic just can't exploit it. This ****validates the Gen-0 ASFormer-over-WASB-features path****, shows the

50 hand-built feature stack the study calls for is the right lever, and confirms the bottleneck is ****data + per-video**

51 calibration (n?5 matches), not method.** The prototype is the seed of Gen-0, and the n=2?n=3 trend (learned mean

52 +0.24, overtaking the heuristic) is the empirical green light.

53

54 **## Platform ranking (privacy + cost lens; the fine-tune is a sub-24h, \$10?50 job ? privacy/ergonomics dominate)**

55

56 1. ****Modal ? PRIMARY.**** Strongest ***written*** no-access contract (verbatim: "Modal will never access? the inputs/outputs to your Functions, or any data you store"), ?7-day TTL, SOC2-II, HIPAA-BAA; per-second scale-to-zero matches retrain-on-each-batch. A100 80 GB ~\$2.50/hr, H100 ~\$3.95/hr (corrected down from cited). Privacy is the load-bearing reason, not price.

57 2. ****RunPod (Secure Cloud tier ONLY) ? FALLBACK / Option-B scale-up.**** Cheapest vetted-DC self-serve; A100 80 GB ~\$1.39?1.49/hr. Community/P2P tier is UNVETTED ? ****never**** touches consented video. Confirm AES-256/DPA at booking.

58 3. ****Vertex AI / GCP ? ON BENCH.**** Only strong-privacy option with an India region, but it's ****capacity-gated**** (asia-south1 high-end GPU is invitation-only, H100-class A3 not A100) and ****tuning Gemini yields NO portable on-device weights**** ? only tuning an Apache-2.0 base in Model Garden does. Use only if India residency-at-rest is mandated.

59 4. ****Self-hosted RTX 2070 Super 8 GB**** ? Gen-0 training + ALL inference + CI only. ****Cannot**** QLoRA a 7B video VLM (~20?24 GB). Best privacy (nothing leaves the box).

60 5. ****Fireworks**** (managed, contractual no-train + ZDR) usable; ****Together BLOCKED**** until a signed DPA + explicit no-train clause. AWS SageMaker / Azure ML = heavy for a solo founder; India-residency only.

61

62 **## Label reuse ? one corpus, three transcoders (the actual moat)**

63

64 Promote the rally CSVs from eval-only files to ****one model-agnostic corpus****: JSONL, one row per

65 `(video\_id, rally\_ordinal)` = `{start, end, ending\_reason, sport, source`
(human|pseudo|vendor),

66 annotator/provenance, consent/opt-in, label_type, split (BY MATCH not clip),
trajectory_ref}`. The

67 provenance/consent fields are the ****enforcement point**** for the legal guardrails. Three deterministic

68 transcoders read it, zero re-labeling:

69 1. ****Heuristic/LogReg view**** ? already exists
(`backend/eval/training.py::label\_candidates` via `metrics.match\_intervals`).

70 2. ****TAL view**** ? ActivityNet JSON (`segment:[start,end]`, `label:'rally'`); our seconds map 1:1.

71 3. ****VLM clip?QA view**** ? "list every rally [start,end] in this clip"; `ending\_reason` ? a second QA task.

72

73 ****One scorer spine:**** reuse `metrics.match\_intervals` for heuristic-labeling, TAL eval, AND the Gen-2 RFT

74 reward ? no metric drift. ****Leakage guard:**** split ****by match, never by clip****. ****Amplification (all**

75 **guardrail-clean):**** open-teacher pseudo-labels (WASB/own-model, confidence-gated ? never paid-LLM),

76 boundary-preserving temporal augmentation, active learning to spend the Roora budget on

```

highest-uncertainty
77     matches. ? **The new partial-label windowing**
(`eval_partial_labels.restrict_to_window`) must flow into the
78     TAL/VLM views so prefix-labeled videos aren't mis-scored (a corpus that's partially
tagged is the norm).
79
80     ## Licensing & privacy guardrails (all respected; verify per-checkpoint)
81
82     - **Bases: Apache-2.0/MIT only.** APPROVED: Qwen2.5-VL-7B/32B (Apache-2.0),
Qwen3-VL-4B/8B (Apache-2.0),
83     Gemma 4 E2B/E4B/12B-Unified (Apache-2.0 + native video/audio), InternVL3 ?38B
(MIT/Apache), ASFormer/
84     ActionFormer/TriDet (MIT). **REJECTED:** Gemma 3 (viral Gemma Terms + PUP),
Qwen2.5-VL-3B (NC research),
85     Qwen-72B/InternVL3-78B (100M-MAU cap), Llama Vision (700M-MAU), DAMO VideoLLaMA3
weights (NC + paid-LLM-trained),
86     TimeChat/VTimeLLM/Momentor (NC/no-license = method-reference only), VideoMAE weights
(CC-BY-NC), `yabufarha/ms-tcn`
87     (Commons Clause ? use `sj-li/MS-TCN2` MIT). ? **Gemma 4 size names =
E2B/E4B/26B-A4B/31B/12B-Unified ? there is no plain "4B".**
88     - **No paid-LLM training data** ? Gemini/OpenAI/Anthropic = inference/eval/teacher-
reference only, **never** a
89     training signal. Gemini silver labels stay out of every transcoder.
90     - **Exclude minors by default + separate training opt-in;** per-user adapters trained
only on that user's
91     consented video (deletion = drop the adapter). Enforced at the corpus-membership layer
before transcoding.
92     - **Privacy lever = local inference** (Gen-0 ~1.1M params; Gen-2 4B quantized). Any
cloud *training* needs a
93     signed DPA + no-train assurance; route only owned/consented footage, never user
uploads.
94
95     ## Cost (spend on labels, not GPUs)
96
97     - **Gen-0:** ~$0 (local 2070 Super; trains in minutes; ~4.5 GB VRAM).
98     - **Gen-2 LoRA:** ~$5?40/run on a rented 24?48 GB GPU (RunPod cheapest; **VRAM/token
cost on real 1080p/60fps
99     clips is UNVERIFIED ? 24 GB is a floor, re-benchmark**). Managed (Fireworks/Together)
~$2?30 *if* text-priced
100     (video tokens can balloon). RFT = compute-heavy (multiple rollouts/sample).
101     - **Inference crossover (the unanchored number):** self-hosted 7B beats Gemini
(~$0.05/10-min clip) **only**
102     above sustained QA volume; below it, **paid Gemini is cheaper** ? paid-primary now,
owned-primary later.
103     **Benchmark $/call before committing** (COMMERCIALIZATION C7).
104
105     ## Open risks & critic caveats (read before acting)
106
107     - **?? WASB checkpoint provenance (C3) propagates.** The study calls WASB "MIT," but
that's the *code*; the
108     distributed `wasb_badminton_best.pth.tar` is **academic-data-trained, provenance
unresolved** (COMMERCIALIZATION
109     C3, still open). Every Gen-0/1 plan uses WASB features as input AND as a pseudo-label
teacher ? a "clean" owned
110     model trained on provenance-unclear features **does not launder the provenance.**
ROADMAP Phase-4 "train own WASB
111     weights" is a **prerequisite, not an aside.**
112     - **Gemma 4 "clean + native-video" rests on post-cutoff secondary sources** ? keep AMBER
until a raw HF LICENSE
113     revision is verified, given the Gemma-3 rejection. Its timestamp-emission recipe is
less documented than Qwen's.
114     - **Existing distilled model was trained on Gemini silver labels** ? the (correct) eval-
only policy now **forbids
115     this; that pipeline must be purged of Gemini-derived training labels** to comply.
116     - **Policy ratifications needed:** guardrail literally says "Apache-2.0 only" but the

```

approved list includes MIT ?

117 ratify "Apache-2.0/MIT". The MAU-cap rejections (72B/78B) may be over-conservative at our scale.

118 - **On-device privacy claim is unbenchmarked** ? no tokens/sec for a quantized 4B VLM on 8 GB Turing at

119 1080p/60fps. **Measure before relying on it.**

120 - **Amplification is load-bearing for Gen-2** ? 128 raw rallies can't reach the ~5k?20k clip?QA band alone; if

121 augmentation + QA-multiplexing + pseudo-labels underdeliver, Option B is under-fed. Amateur 60fps may need more

122 labels than broadcast-trained literature implies.

123 - **Dropped option worth keeping:** InternVideo2 encoder (MIT) + light grounding head ? the *only* data-realistic

124 owned-video experiment available at n?128 today. And: a **one-time ~\$1?2k 24 GB local card** (e.g. used 3090)

125 would preserve the privacy lever for *training* and dissolve most of the cloud-DPA analysis ? uncoded here.

126

127 ## Open questions (need the GPU/WSL machine-side suite ? not runnable in this dev container)

128

129 1. Does ASFormer/ActionFormer over the WASB feature stack beat the **honest ~0.44 LOVO** bar at n?5 matches?

130 (Today's prototype says the per-frame AUC is there ? 0.84 ? so this is now *likely*; confirm at segment level.)

131 2. Real VRAM + token cost of fine-tuning Qwen3-VL-4B / Qwen2.5-VL-7B on our actual clip?QA frame counts?

132 3. Working Gemma-4 timestamp-emission recipe + exact checkpoint (E4B vs 12B-Unified)?

133 4. Measured self-hosted-7B \$/call + latency vs Gemini on the target workload (the owned-model ROI crossover)?

134 5. Will Together/Lambda/Azure sign a DPA with explicit no-train for consented video, India-configurable?

135 6. Realistic clip?QA yield from 128 rallies via augmentation + QA-multiplexing + pseudo-labels?

136

137 ## Sequencing (owner directive, non-negotiable)

138

139 This is a **design blueprint only.** It unblocks PLATFORM **P5** but starts **no implementation** until P0?P4

140 scaffolding (async jobs, storage/compute backends) is rock-solid AND the golden corpus is healthy. Paid Gemini

141 stays the always-ready primary/fallback. The only items safe to act on now ? **and even these need owner

142 sign-off before code** ? are (a) the model-agnostic corpus format and (b) growing n?2 ? n?5 matches (a small

143 Roora batch), which unblocks honest tuning immediately and is the **highest-ROI spend (labels, not GPUs).**

144

145 *Full agent transcript + 35 sources: workflow `wf\_49aab888-e10` (this session).*

146

8. assistant (2026-06-09T11:25:54.277Z)

These two docs are extremely rich and directly relevant. The existing study already lands on a three-generation plan (TAS head now ? VLM later). My job is to validate/challenge the core fork ? can a fine-tuned VLM hit high P/R on temporal localization vs specialized TAL heads ? with fresh 2026 web evidence, and produce a recommendation. Let me now fan out web research on the key empirical questions.

9. user (2026-06-09T11:26:02.824Z)

Web search results for query: "video LLM temporal grounding 2025 2026 benchmark Charades-STA mAP vs specialized temporal action localization ActionFormer"

Links: [{"title": "TimeRefine: Temporal Grounding with Time Refining Video

```
LLM", "url": "https://arxiv.org/html/2412.09601v1"}, {"title": "TEMPURA: Temporal Event Masked Prediction and Understanding for Reasoning in Action", "url": "https://arxiv.org/pdf/2505.01583"}, {"title": "Temporal Grounding of Activities using Multimodal Large Language Models", "url": "https://arxiv.org/pdf/2407.06157"}, {"title": "Agentic Video Intelligence: A Flexible Framework for Advanced Video Exploration and Understanding", "url": "https://arxiv.org/pdf/2511.14446"}, {"title": "VideoTemp-o3: Harmonizing Temporal Grounding and Video Understanding in Agentic Thinking-with-Videos", "url": "https://arxiv.org/html/2602.07801v3"}, {"title": "TimeExpert: An Expert-Guided Video LLM for Video Temporal Grounding", "url": "https://openaccess.thecvf.com/content/ICCV2025/papers/Yang_TimeExpert_An_Expert-Guided_Video_LLM_for_Video_Temporal_Grounding_ICCV_2025_paper.pdf"}, {"title": "TEMPORAL GROUNDING VIDEO LLM VIA CAUSAL ...", "url": "https://openreview.net/pdf/7a9214347999784f640ab6597bc776dbd5999cea.pdf"}, {"title": "Enrich and Detect: Video Temporal Grounding with Multimodal LLMs", "url": "https://www.openaccess.thecvf.com/content/ICCV2025/papers/Pramanick_Enrich_and_Detect_Video_Temporal_Grounding_with_Multimodal_LLMs_ICCV_2025_paper.pdf"}]
```

Based on the search results, here's what I found regarding video LLM temporal grounding benchmarks and recent developments:

Recent Video LLM Temporal Grounding Research (2025-2026)

Recent work uses Charades-STA and ActivityNet-MR benchmarks to test the model's ability to localize temporal segments corresponding to text captions, with performance measured using mIoU (mean Intersection over Union) and other metrics like R@IoU (Recall at specific IoU thresholds).

Video LLM Performance on Charades-STA

Several recent methods have been evaluated on this benchmark:

- The AVI method achieves the highest R@0.3 score (88.6%) and mIoU (60.0%), surpassing previous best methods including the RL-trained VITAL on Charades-STA
- TimeRefine achieves 5.0% mIoU improvements on the Charades-STA dataset
- ED-VTG achieves strong improvements over previous generalist models, and performs comparably to task-specific expert models

Comparison with Specialized Methods

ED-VTG surpasses all specialist models on the Charades-CD-00D dataset by a substantial margin, setting a new state-of-the-art. This suggests that newer video LLM approaches are competitive with or exceeding specialized temporal action localization methods.

However, the search results don't contain specific information comparing ActionFormer directly against these video LLM approaches on the same metrics.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

10. user (2026-06-09T11:26:07.304Z)

Web search results for query: "VLM temporal video grounding fine-tuned timestamp emission accuracy limitations 2025 survey"

Links: [{"title": "A Survey on Video Temporal Grounding with Multimodal Large Language Model", "url": "https://arxiv.org/html/2508.10922v1"}, {"title": "TAR-TVG: Enhancing VLMs with Timestamp Anchor-Constrained Reasoning for Temporal Video Grounding", "url": "https://arxiv.org/pdf/2508.07683"}, {"title": "Temporal Grounding as a Learning Signal for Referring Video Object Segmentation", "url": "https://arxiv.org/pdf/2508.11955"}, {"title": "Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding", "url": "https://xuboshen.github.io/Time-R1/"}, {"title": "Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding", "url": "https://arxiv.org/pdf/2503.13377"}, {"title": "VideoExpert: Augmented LLM for"}]

Temporal-Sensitive Video

Understanding", "url": "https://arxiv.org/html/2504.07519v1"}, {"title": "Know-Show: Benchmarking Video-Language Models on Spatio-Temporal Grounded Reasoning", "url": "https://arxiv.org/pdf/2512.05513"}, {"title": "Qwen3-VL Technical Report", "url": "https://arxiv.org/pdf/2511.21631"}]]

Based on the search results, here's what I found regarding VLM temporal video grounding with timestamp emission and accuracy limitations:

Key Survey and Recent Findings

Recent advancement in video temporal grounding (VTG) has significantly enhanced fine-grained video understanding, primarily driven by multimodal large language models (MLLMs), with VTG-MLLMs gradually surpassing traditional fine-tuned methods.

Timestamp and Accuracy Limitations

A major limitation regarding timestamp accuracy has been identified: In Qwen2.5-VL's time-synchronized approach, tying temporal position IDs directly to absolute time produces excessively large and sparse temporal position ids for long videos, degrading the model's ability to understand long temporal contexts, and effective learning requires extensive and uniformly distributed sampling across various frame rates.

Models capture coarse temporal regions rather than accurate event boundaries, and current Video-LMs exhibit relative temporal awareness but struggle with absolute fine-grained temporal grounding.

Performance Gaps

A significant performance gap exists between simpler and more complex scenarios in VLM temporal localization capabilities, with most VLMs achieving above 65% Top-1 accuracy on simpler sets but dropping to around 50% on more challenging sets.

Training Approaches

Fine-tuning, pretraining, and training-free approaches each have unique trade-offs in generalization, task specialization, and supervision requirements. Recent work like Time-R1 addresses generalization limitations through post-training with reinforcement learning.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-06-09T11:26:12.009Z)

This is a key finding: the 2025-2026 literature shows VLMs for temporal grounding are now "gradually surpassing traditional fine-tuned methods" on Charades-STA, but with a critical caveat ? they capture "coarse temporal regions rather than accurate event boundaries" and "struggle with absolute fine-grained temporal grounding." That tension is central to the high-P/R question. Let me dig deeper on the precision numbers and the specialized-head ceiling, plus the multisport/single-model angle.

12. user (2026-06-09T11:26:20.139Z)

Web search results for query: "ActionFormer ActivityNet THUMOS14 mAP state of the art 2024 2025 temporal action localization accuracy"

Links: [{"title": "[2202.07925] ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/html/2202.07925"}, {"title": "(PDF) ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.researchgate.net/publication/358655346_ActionFormer_Localizing_Moments_of_Actions_with_Transformers"}, {"title": "TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://arxiv.org/html/2512.01298"}, {"title": "Benchmarking Data Efficiency

and Computational Efficiency", "url": "https://cveu.github.io/2023/papers/11.pdf"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/pdf/2202.07925"}, {"title": "Detecting Informative Channels: ActionFormer", "url": "https://arxiv.org/html/2505.20739v1"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136640485.pdf"}, {"title": "A Multi-Modal Transformer Network for Action Detection", "url": "https://arxiv.org/pdf/2305.19624"}, {"title": "arXiv:2504.00647v1 [cs.CV] 1 Apr 2025", "url": "https://arxiv.org/pdf/2504.00647"}, {"title": "Spatiotemporal Event Graphs for Dynamic Scene Understanding", "url": "https://arxiv.org/pdf/2312.07621"}]

Based on the search results, here's what I found regarding ActionFormer and state-of-the-art temporal action localization performance on THUMOS14 and ActivityNet:

ActionFormer Performance

ActionFormer achieved 65.5% mAP@0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points, and reaches an average mAP of 36.6% on ActivityNet 1.3. At the time of its proposal, the model reached state-of-the-art performance on the THUMOS'14 dataset obtaining an average mAP of 66.8%.

Recent State-of-the-Art (2024-2025)

More recent work has surpassed ActionFormer. TBT-Former establishes a new state-of-the-art on the THUMOS14 benchmark. It achieves an average mAP of 68.0%, surpassing the strong ActionFormer baseline by 1.2 absolute points. This result is from December 2025, representing the latest state-of-the-art in the field.

Summary

- **ActionFormer**: 66.8% average mAP on THUMOS14 (original proposal in 2022)
- **TBT-Former**: 68.0% average mAP on THUMOS14 (December 2025) - current state-of-the-art

The field has continued to progress beyond ActionFormer's original results, with newer methods achieving improved performance on these temporal action localization benchmarks.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

13. user (2026-06-09T11:26:22.115Z)

Web search results for query: "Time-R1 Qwen2.5-VL temporal grounding R@0.7 high IoU recall precision Charades-STA mIoU numbers"

Links: [{"title": "GitHub - www-Ye/Time-R1: R1-like Video-LLM for Temporal Grounding · GitHub", "url": "https://github.com/www-Ye/Time-R1"}, {"title": "Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding", "url": "https://arxiv.org/html/2503.13377v2"}, {"title": "Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding", "url": "https://xuboshen.github.io/Time-R1/"}, {"title": "TimeLens: Rethinking Video Temporal Grounding with Multimodal LLMs", "url": "https://arxiv.org/html/2512.14698"}, {"title": "xiaomi-research/time-r1 | DeepWiki", "url": "https://deepwiki.com/xiaomi-research/time-r1"}, {"title": "Agentic Video Intelligence: A Flexible Framework for Advanced Video Exploration and Understanding", "url": "https://arxiv.org/pdf/2511.14446"}, {"title": "EvoGround: Self-Evolving Video Agents for Video Temporal Grounding", "url": "https://arxiv.org/html/2605.13803"}, {"title": "Qwen2.5-VL: Advanced Vision-Language Model", "url": "https://www.emergentmind.com/topics/qwen2-5-vl-model"}, {"title": "Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding", "url": "https://arxiv.org/pdf/2503.13377"}]

Based on the search results, I found detailed information about Time-R1's temporal grounding

performance on Charades-STA:

On Charades-STA, Time-R1 achieves R1@0.3 of 82.8%, R1@0.5 of 72.2%, R1@0.7 of 50.1%, and mIoU of 60.9% in the fine-tuned setting (marked with *). On ActivityNet, Time-R1 (referred to as TimeZero in the results) achieves R1@0.3 of 83.3%, R1@0.5 of 72.5%, and R1@0.7 of 47.9%.

Time-R1 is built on the Qwen2.5-VL-7B base model. The method introduces a reasoning-guided post-training framework via RL with verifiable reward to enhance capabilities on the temporal video grounding task.

These results demonstrate strong performance on the Charades-STA benchmark, with particularly notable improvements at the high IoU threshold (R@0.7), where the model achieves 50.1% recall.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. assistant (2026-06-09T11:26:28.221Z)

Now I have concrete numbers. Key data point: even a strong RL-post-trained VLM (Time-R1 on Qwen2.5-VL-7B) reaches only R@0.7 = 50.1% on Charades-STA ? meaning at a strict IoU=0.7, half the predicted moments are still misaligned. That is NOT "very high precision/recall" at strict boundaries. Specialized TAL (ActionFormer) reaches ~66.8% mAP on THUMOS14 (a harder multi-instance setting). Let me now probe the multisport/single-model feasibility and the sports-specific TAL question, plus the conversational-QA architecture (RAG-over-events vs single model).

15. user (2026-06-09T11:26:38.241Z)

Web search results for query: "video question answering long video agentic retrieval over event index vs end-to-end VLM 2025 architecture"

Links: [{"title": "Video Language Models Overview", "url": "https://www.emergentmind.com/topics/video-language-models-vlm"}, {"title": "Empowering Agentic Video Analytics Systems with Video Language Models", "url": "https://arxiv.org/html/2505.00254v3"}, {"title": "LongVideoAgent: Multi-Agent Reasoning with Long Videos", "url": "https://arxiv.org/pdf/2512.20618"}, {"title": "Think, Then Verify: A Hypothesis-Verification Multi-Agent Framework for Long Video Understanding", "url": "https://arxiv.org/pdf/2603.04977"}, {"title": "MemoryCard: Topic-Aware Multi-Modal Clue Compression for Long-Video Question Answering", "url": "https://arxiv.org/html/2606.05917"}, {"title": "Active Video Perception: Iterative Evidence Seeking for Agentic Long Video Understanding", "url": "https://arxiv.org/pdf/2512.05774"}, {"title": "LongVidSearch: An Agentic Benchmark for Multi-hop Evidence Retrieval Planning in Long Videos", "url": "https://arxiv.org/pdf/2603.14468"}]

I found relevant research on video question answering with different architectural approaches. Here are the key findings:

Agentic Retrieval-Based Systems

AVAS builds an Event Knowledge Graph indexing long streams, supporting retrieval-augmented agentic reasoning and factual QA. Recent studies attempting to extend the context window of VLMs remain inadequate for processing video sources spanning hundreds of hours, while retrieval-augmented generation (RAG) frameworks aim to address similar limitations by first retrieving relevant frames from massive contents and then generating final answers.

Challenges with Long Videos

These approaches still face significant challenges in handling the video modality, leading to both reduced analytics accuracy and substantial computational overhead. Recent trends emphasize retrieval-aware reasoning and efficient tokenization for long videos, with agentic planners that iteratively gather evidence, motivating long-form VideoQA systems that selectively retrieve segments under a limited context budget.

Emerging Approaches

ProgGen uses VLM/LLM to synthesize interpretable world state extractors, next-state physics routines, and RGB renderers for programmatic video prediction. Additionally, MemoryCard performs intensive reading over video sessions to generate event-level video gists with VLM-generated topics and aligned utterances, rendering these into a unified Memory Card.

The research indicates that agentic retrieval systems with event-based indexing are emerging as alternatives to end-to-end VLM approaches for handling long-form video question answering efficiently.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

16. user (2026-06-09T11:26:38.822Z)

Web search results for query: "sports temporal action localization racket sports badminton tennis table tennis single model multi-sport generalization 2025"

Links: [{"title": "Bridging the Gap: Doubles Badminton Analysis with Singles-Trained Models", "url": "https://arxiv.org/html/2508.13507v1"}, {"title": "SportR: A Benchmark for Multimodal Large Language Model Reasoning in Sports", "url": "https://arxiv.org/html/2511.06499v2"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis", "url": "https://arxiv.org/html/2511.17045v2"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis", "url": "https://arxiv.org/pdf/2511.17045"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified", "url": "https://ojs.aaai.org/index.php/AAAI/article/view/37362/41324"}, {"title": "Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v3"}, {"title": "Survey of Action Recognition, Spotting and Spatio-Temporal Localization in Soccer -- Current Trends and Research Perspectives", "url": "https://arxiv.org/pdf/2309.12067"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/html/2502.21085v2"}]

I found relevant search results related to your query about sports temporal action localization and multi-sport generalization in racket sports. Here are the key findings:

RacketVision: A Multiple Racket Sports Benchmark

RacketVision is a multiple racket sports benchmark for unified ball and racket analysis that expands the range of sports types for unified model training, aiming to uncover shared priors across racket sports, specifically comprising 1,672 video clips spanning badminton, tennis, and table tennis.

The benchmark progressively proposes three tasks with corresponding annotations: beyond the existing ball tracking task, RacketVision defines racket keypoints and supports a novel racket pose estimation task, and further proposes an integrative task of ball trajectory prediction.

Multi-Sport Generalization Results

Comparing the best multi-sport model (MS-TrackNetV3) with the best single-sport model (TrackNetV3) reveals a clear trend of improved generalization, with the multi-sport model boosting mAP by 19.2% in Tennis and 14.6% in Badminton. However, this broad generalization sometimes comes at the cost of hyper-specialized precision on a single sport.

Additional Related Work

Recent datasets have embraced precise event detection formulations, including datasets for tennis and table tennis. Additionally, ProtoGCN, a method based on custom GCN and multi-scale TCN with Class-Specific Contrastive Loss, achieved strong performance on several public datasets

in skeleton-based action recognition and was published at CVPR 2025.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

17. assistant (2026-06-09T11:26:50.247Z)

Excellent ? RacketVision (AAAI 2026) is a direct hit: a unified multi-sport (badminton/tennis/TT) model improves generalization (+19.2% mAP tennis, +14.6% badminton) but "broad generalization sometimes comes at the cost of hyper-specialized precision." And the QA-architecture literature strongly favors event-knowledge-graph / RAG-over-index (AVAS) over end-to-end VLM for the conversational use case. Let me fetch the RacketVision and the VTG survey papers directly to extract precise evidence, and do one more search on data/compute requirements for fine-tuning a video VLM for grounding.

18. user (2026-06-09T11:26:59.993Z)

Key Findings on VLM-Based Video Temporal Grounding

(1) Performance Claims vs. Specialized Methods

The survey states that VTG-MLLMs "are gradually surpassing traditional fine-tuned methods" and "achieve competitive performance." However, the experimental data in Table II reveals a more nuanced picture: while some pretrained VLM approaches show strength in mid-range metrics, they don't decisively outperform all baselines, particularly at higher IoU thresholds.

(2) Benchmark Performance Numbers

****Charades-STA (Zero-shot):****

- Best method: Time-R1 (PT) - R@0.5: 60.8%, R@0.7: 35.3%
- Strong competitor: TFGVTG (TF) - R@0.5: 50.0%, R@0.7: 24.3%

****ActivityNet-Captions (Zero-shot):****

- Best reported: Ground-VideoLLM (PT) - mIoU: 36.1%, R@0.5: 30.3%
- Comparable: TPE-VLLM (PT) - mIoU: 36.1%, R@0.5: 35.4%

(3) Stated Limitations

The document explicitly identifies challenges: "standard MLLMs often struggle to capture fine-grained temporal dependencies" and face difficulty with "precise temporal boundary prediction." The survey notes these models grapple with "bridging semantic gaps" and exhibit "limited generalization capabilities" in certain contexts.

(4) Data and Compute Requirements

The survey indicates pretraining approaches demand "high computational cost" and "substantial effort required to construct large, high-quality temporal datasets." Specific dataset scales mentioned include Moment-10M and E.T. Instruct 164K, though explicit compute metrics aren't provided in the excerpt.

19. user (2026-06-09T11:27:02.501Z)

Web search results for query: "fine-tuning video VLM temporal grounding data requirements number of samples GRPO RL cold-start SFT compute cost 2025"

Links: [{"title": "Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding", "url": "https://xuboshen.github.io/Time-R1/"}, {"title": "Reinforcing Video Reasoning Segmentation to Think Before It Segments", "url": "https://arxiv.org/html/2508.11538"}, {"title": "Time-R1: Post-Training Large Vision Language Model for Temporal Video Grounding", "url": "https://arxiv.org/html/2503.13377v2"}, {"title": "STVG-R1: Incentivizing"}]

Instance-Level Reasoning and Grounding in Videos via Reinforcement Learning", "url": "https://arxiv.org/pdf/2602.11730"}, {"title": "Datasets and Recipes for Video Temporal Grounding via Reinforcement Learning - ACL Anthology", "url": "https://aclanthology.org/2025.emnlp-industry.66/"}, {"title": "Zoom-Zero: Reinforced Coarse-to-Fine Video Understanding via Temporal Zoom-in", "url": "https://arxiv.org/pdf/2512.14273"}, {"title": "MUSEG: Reinforcing Video Temporal Understanding via Timestamp-Aware Multi-Segment Grounding", "url": "https://arxiv.org/pdf/2505.20715"}, {"title": "Video-OPD: Efficient Post-Training of Multimodal Large Language Models for Temporal Video Grounding via On-Policy Distillation", "url": "https://arxiv.org/pdf/2602.02994"}]

Based on the web search results, I found several recent papers (2025) addressing video VLM temporal grounding with GRPO-based RL training and data requirements:

Key Findings:

Time-R1 Framework: Time-R1 is a reasoning-enhanced post-training framework via RL with verifiable rewards, where the LLM first generates chain-of-thought descriptions and then predicts timestamps, optimized using Generalized Reinforcement Policy Optimization (GRPO) with a novel reward function incorporating both structured template reward and timestamp-aware tIoU reward.

Data Efficiency: Unlike prior methods that rely on large-scale supervised fine-tuning, Time-R1 leverages a verifiable reward signal to unlock strong temporal reasoning from pretrained models using limited data.

Cold-Start SFT Strategy: To ensure stable reasoning and reduce hallucinations, a cold-start approach is adopted to generate chain-of-thought with video captions. Additionally, the approach leverages high-quality curated cold-start data for SFT initialization, followed by difficulty-controlled RL to further enhance temporal localization and reasoning abilities.

Reinforcement Fine-Tuning: TimeRFT is a reinforcement fine-tuning strategy with dynamic hard sampling, which mines hard samples on a curated dataset and progressively selects low-IoU samples for multi-epoch training.

Computational Efficiency Challenges: GRPO relies on multiple on-policy rollouts to obtain low-variance reward estimates, and in video understanding where each rollout conditions on long visual contexts, the cost of generating trajectories scales prohibitively with the number of rollouts, incurring substantial computational overhead during training.

However, the search results don't contain specific numbers for exact sample counts or detailed compute cost measurements for 2025.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

20. assistant (2026-06-09T11:27:09.997Z)

Critical nuance: the survey's *zero-shot* numbers are much lower (Time-R1 ZS R@0.7=35.3%) than the *fine-tuned* numbers (R@0.7=50.1%). This confirms: to get a VLM to even ~50% R@0.7, you must fine-tune on the target domain, and even then strict-boundary recall caps around half. Let me fetch the RacketVision paper for the multi-sport evidence and do a final search on the data scale needed plus the "GRPO needs verifiable reward" angle which maps directly to this project's tIoU-vs-golden reward design.

21. user (2026-06-09T11:27:18.617Z)

RacketVision: Key Findings Summary

(1) What is RacketVision and Which Sports?

RacketVision is "a novel dataset and benchmark for advancing computer vision in sports

analytics, covering table tennis, tennis, and badminton." It comprises 1,672 video clips (435,179 frames) with annotations for both ball positions and racket poses?the first dataset to provide large-scale racket keypoint annotations alongside ball tracking data.

(2) Multi-Sport vs Single-Sport Performance

The research demonstrates significant gains from unified multi-sport training:

- **Tennis mAP**: Multi-sport model achieved 81.9% versus 68.7% for single-sport (19.2% improvement)
- **Badminton mAP**: 83.1% versus 72.5% (14.6% improvement)
- **Precision tradeoff**: Single-sport models occasionally retained slight edges in pinpoint localization accuracy, suggesting "training on diverse data forces the model to learn more robust features" at modest localization precision costs.

(3) Single Shared Model Effectiveness

Yes?the multi-sport model (MS-TrackNetV3) substantially outperformed sport-specific variants. The research notes that unified training enhanced "model generalization" across all three sports, though with the localization precision tradeoffs mentioned above.

(4) VLM/MLLM Performance

The document contains **no discussion** of Vision Language Models or Multimodal Large Language Models performing these tasks.

(5) Licensing and Availability

Not specified. The paper provides no information about dataset licensing, access restrictions, or repository locations.

22. user (2026-06-09T11:27:21.845Z)

Web search results for query: "datasets and recipes" video temporal grounding reinforcement learning EMNLP 2025 sample efficiency how many training examples"

Links: [{"title": "GitHub - zjuruizhechen/TVG-R1: [EMNLP 2025 Industry] Datasets and Recipes for Video Temporal Grounding via Reinforcement Learning · GitHub", "url": "https://github.com/zjuruizhechen/TVG-R1"}, {"title": "Datasets and Recipes for Video Temporal Grounding via Reinforcement Learning - ACL Anthology", "url": "https://aclanthology.org/2025.emnlp-industry.66/"}, {"title": "[2507.18100] Datasets and Recipes for Video Temporal Grounding via Reinforcement Learning", "url": "https://arxiv.org/abs/2507.18100"}, {"title": "Datasets and Recipes for Video Temporal Grounding via ...", "url": "https://aclanthology.org/2025.emnlp-industry.66.pdf"}, {"title": "VideoTG-R1: Boosting Video Temporal Grounding via Curriculum Reinforcement Learning on Reflected Boundary Annotations", "url": "https://arxiv.org/pdf/2510.23397"}, {"title": "arXiv Preprint Datasets and Recipes for Video Temporal Grounding via", "url": "https://arxiv.org/pdf/2507.18100"}, {"title": "Datasets and Recipes for Video Temporal Grounding via Reinforcement Learning", "url": "https://arxiv.org/html/2507.18100v1"}, {"title": "Learning Transferable Temporal Primitives for Video Reasoning via Synthetic Videos", "url": "https://arxiv.org/pdf/2603.17693"}, {"title": "GitHub - WHB139426/Grounded-Video-LLM: [EMNLP 2025 Findings] Grounded-VideoLLM: Sharpening Fine-grained Temporal Grounding in Video Large Language Models · GitHub", "url": "https://github.com/whb139426/grounded-video-llm"}]

Based on the search results, I found information about the paper "Datasets and Recipes for Video Temporal Grounding via Reinforcement Learning" from EMNLP 2025.

Key Findings:

Paper Overview:

This repository presents datasets and recipes to enhance temporal grounding and reasoning capabilities in large vision-language models (LVLMs) for video temporal grounding tasks.

****Training Datasets:****

The framework performs supervised fine-tuning (SFT) with curated high-quality cold-start data to initialize the base model, followed by reinforcement learning (RL), using a Coldstart-13K dataset for SFT-based coldstart and an RL-18K subset.

****Training Examples:****

The authors release full 56K Chain-of-Thought temporal video grounding annotation by Gemini-2.5-Pro. This indicates the total number of training examples available in their dataset.

****Key Research Contributions:****

The work conducts an in-depth analysis of training strategies and dataset curation, highlighting the importance of both high-quality cold-start data and difficulty-controlled RL.

****Publication Details:****

The work was authored by Ruizhe Chen, Tianze Luo, Zhiting Fan, and colleagues, presented at the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track in Suzhou, China.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

23. assistant (2026-06-09T11:27:29.078Z)

Now I have a strong evidence base. Two more important data points to nail down:

1. The data-scale recipe: SFT cold-start ~13K + RL ~18K examples is the proven recipe for VLM grounding ? and notably the public CoT annotations were generated by Gemini-2.5-Pro (which this project's guardrail forbids using). That's a concrete corroboration of the "no-paid-LLM-training narrows the path" tension.
2. Let me confirm the "VLM struggles at strict IoU but the head wins at strict boundaries" and check whether a hybrid (VLM proposes, head refines, or head detects + LLM does QA) is the emerging consensus.

24. user (2026-06-09T11:27:38.761Z)

Web search results for query: "hybrid video temporal grounding VLM coarse localization plus specialized boundary refinement head 2025 two-stage"

Links: [{"title": "TimeRefine: Temporal Grounding with Time Refining Video LLM", "url": "https://arxiv.org/pdf/2412.09601"}, {"title": "Grounded-VideoLLM: Sharpening Fine-grained Temporal ...", "url": "https://aclanthology.org/2025.findings-emnlp.50.pdf"}, {"title": "TimeRefine: Temporal Grounding with Time Refining Video LLM", "url": "https://arxiv.org/html/2412.09601v1"}, {"title": "TimeExpert: An Expert-Guided Video LLM for Video Temporal Grounding", "url": "https://openaccess.thecvf.com/content/ICCV2025/papers/Yang_TimeExpert_An_Expert-Guided_Video_LLM_for_Video_Temporal_Grounding_ICCV_2025_paper.pdf"}, {"title": "1 A Survey on Video Temporal Grounding with Multimodal Large Language Model", "url": "https://arxiv.org/pdf/2508.10922"}, {"title": "Universal Video Temporal Grounding with Generative Multi ...", "url": "https://papers.neurips.cc/paper_files/paper/2025/file/5d2e24df9cfaad3189833b819c40b392-Paper-Conference.pdf"}, {"title": "GitHub - iLearn-Lab/TPAMI26-Awesome-MLLMs-for-Video-Temporal-Grounding: Latest Papers, Codes and Datasets on VTG-LLMs. .", "url": "https://github.com/ki-lw/Awesome-MLLMs-for-Video-Temporal-Grounding"}, {"title": "Enrich and Detect: Video Temporal Grounding with Multimodal LLMs", "url": "https://arxiv.org/html/2510.17023v1"}, {"title": "Universal Video Temporal Grounding with Generative Multi-modal Large Language Models", "url": "https://arxiv.org/html/2506.18883v2"}]

Based on the search results, I found several relevant papers on video temporal grounding with VLMs that employ two-stage approaches with coarse localization and boundary refinement. Here are the key findings:

TimeRefine: Temporal Grounding with Time Refining Video LLM

This model generates a coarse prediction initially and then progressively refines it via temporal offset prediction for more precise temporal localization. TimeRefine proposes a step-by-step refinement paradigm to the LLM-based temporal grounding task, which helps the model learn to self-correct its errors in boundary perception, thus enhancing the Video LLM's temporal understanding ability.

General Two-Stage Approaches in 2025

Recent research shows several methods implementing similar hybrid strategies. Some models perform this as an iterative refinement task, where the model first predicts coarse intervals (e.g., "15.0s to 27.5s") and subsequently refines these estimates by predicting offset adjustments (e.g., "+4.0s and -1.5s"), achieving fine-grained localization through iterative reasoning.

One approach introduces a lightweight perception decoder that specializes in generating precise temporal boundaries, conditioned on a contextual representation, allowing the LLM to focus on language outputs.

Universal Video Temporal Grounding Framework

A universal temporal grounding framework operates in two stages: retrieving the video segment most relevant to the query, then predicting the precise temporal interval within it, where the model first retrieves a coarse relevant segment and then localizes the precise moment within it.

These approaches demonstrate the effectiveness of combining VLM-based coarse localization with specialized refinement heads for achieving fine-grained temporal grounding in 2025.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

25. user (2026-06-09T11:27:45.934Z)

Web search results for query: "Qwen3-VL temporal grounding timestamp localization video benchmark results 2026 improvement over Qwen2.5-VL"

Links: [{"title": "Qwen3-VL | OpenLM.ai", "url": "https://openlm.ai/qwen3-vl/"}, {"title": "GitHub - TencentARC/TimeLens: [CVPR 2026] TimeLens: Rethinking Video Temporal Grounding with Multimodal LLMs · GitHub", "url": "https://github.com/TencentARC/TimeLens"}, {"title": "GitHub - QwenLM/Qwen3-VL: Qwen3-VL is the multimodal large language model series developed by Qwen team, Alibaba Cloud. · GitHub", "url": "https://github.com/qwenlm/qwen3-vl"}, {"title": "qwen3-vl", "url": "https://lmstudio.ai/models/qwen3-vl"}, {"title": "[2511.21631] Qwen3-VL Technical Report", "url": "https://arxiv.org/abs/2511.21631"}, {"title": "Qwen3-VL Technical Report", "url": "https://www.researchgate.net/publication/398026379_Qwen3-VL_Technical_Report"}, {"title": "Qwen3-VL Technical Report [Quick Review]", "url": "https://liner.com/review/qwen3vl-technical-report"}, {"title": "Video Understanding | QwenLM/Qwen3-VL | DeepWiki", "url": "https://deepwiki.com/QwenLM/Qwen3-VL/5.4-video-understanding"}, {"title": "UniversalVTG: A Universal and Lightweight Foundation Model for Video Temporal Grounding", "url": "https://arxiv.org/pdf/2604.08522"}, {"title": "Qwen/Qwen3-VL-8B-Instruct · Hugging Face", "url": "https://huggingface.co/Qwen/Qwen3-VL-8B-Instruct"}]

Let me search for more specific benchmark results comparing Qwen3-VL and Qwen2.5-VL.

Links: [{"title": "VideoAuto-R1: Video Auto Reasoning via Thinking Once, Answering Twice", "url": "https://arxiv.org/pdf/2601.05175"}, {"title": "Video-o3: Native Interleaved Clue Seeking for Long Video Multi-Hop Reasoning", "url": "https://arxiv.org/pdf/2601.23224"}, {"title": "Introduction to Qwen3-VL", "url": "https://debuggercafe.com/introduction-to-qwen3-vl/"}, {"title": "Open-o3-Video: Grounded Video Reasoning with Explicit Spatio-Temporal Evidence", "url": "https://arxiv.org/pdf/2510.20579"}, {"title": "Qwen2.5-VL Technical Report", "url": "https://arxiv.org/pdf/2502.13923"}, {"title": "VAU-R1: Advancing Video Anomaly"}]

Understanding via Reinforcement Fine-Tuning", "url": "https://arxiv.org/pdf/2505.23504"}, {"title": "VideoTemp-o3: Harmonizing Temporal Grounding and Video Understanding in Agentic Thinking-with-Videos", "url": "https://arxiv.org/pdf/2602.07801"}, {"title": "VTimeCoT: Thinking by Drawing for Video Temporal Grounding and Reasoning", "url": "https://arxiv.org/pdf/2510.14672"}, {"title": "Qwen3-VL Technical Report [Quick Review]", "url": "https://liner.com/review/qwen3vl-technical-report"}, {"title": "Video-in-the-Loop: Span-Grounded Long Video QA with Interleaved Reasoning", "url": "https://arxiv.org/pdf/2510.04022"}]

Let me search for more specific benchmark numbers from the Qwen3-VL technical report.

Links: [{"title": "Agentic Video Intelligence: A Flexible Framework for Advanced Video Exploration and Understanding", "url": "https://arxiv.org/pdf/2511.14446"}, {"title": "Qwen3-VL Technical Report", "url": "https://www.researchgate.net/publication/398026379_Qwen3-VL_Technical_Report"}, {"title": "T2SGrid: Temporal-to-Spatial Gridification for Video Temporal Grounding", "url": "https://arxiv.org/pdf/2603.06973"}, {"title": "TempSamp-R1: Effective Temporal Sampling with Reinforcement Fine-Tuning for Video LLMs", "url": "https://arxiv.org/pdf/2509.18056"}, {"title": "[2511.21631] Qwen3-VL Technical Report", "url": "https://arxiv.org/abs/2511.21631"}, {"title": "Training-free Uncertainty Guidance for Complex Visual Tasks with MLLMs", "url": "https://arxiv.org/pdf/2510.00705"}, {"title": "TimeScope: Towards Task-Oriented Temporal Grounding In Long Videos", "url": "https://arxiv.org/pdf/2509.26360"}, {"title": "Universal Video Temporal Grounding with Generative Multi-modal Large Language Models", "url": "https://arxiv.org/pdf/2506.18883"}, {"title": "Invert4TVG: A Temporal Video Grounding Framework with Inversion Tasks Preserving Action Understanding Ability", "url": "https://arxiv.org/pdf/2508.07388"}, {"title": "Weakly-Supervised Multi-Level Attentional Reconstruction Network for Grounding Textual Queries in Videos", "url": "https://arxiv.org/pdf/2003.07048"}]

Based on the search results, I found helpful information about Qwen3-VL's temporal grounding capabilities, though specific benchmark score comparisons between Qwen3-VL and Qwen2.5-VL are limited in the publicly available materials. Here's what I found:

Qwen3-VL Temporal Grounding and Timestamp Localization

Qwen3-VL introduces text-based time alignment for video, evolving from T-RoPE to explicit textual timestamp alignment for more precise temporal grounding. The model features precise text-timestamp alignment for temporal event localization, enabling second-level indexing of video content, moving beyond traditional temporal position embeddings (T-RoPE) to provide grounded, timestamp-aware responses.

Key Architectural Improvements

Qwen3-VL's three key architectural upgrades include an enhanced interleaved-MRoPE for stronger spatial-temporal modeling across images and video, DeepStack integration which leverages multi-level ViT features to tighten vision-language alignment, and text-based time alignment for video, evolving from T-RoPE to explicit textual timestamp alignment for more precise temporal grounding.

Qwen2.5-VL Baseline Performance

For context, on the Charades-STA dataset, Qwen2.5-VL-72B achieves an mIoU score of 50.9, and Qwen2.5-VL-7B achieves 67.9 R@0.3, 50.3 R@0.5, 24.3 R@0.7, and 43.6 mIoU on Charades-STA.

Unfortunately, the search results do not contain specific published benchmark results directly comparing Qwen3-VL's temporal grounding scores to Qwen2.5-VL on standard benchmarks like Charades-STA or ActivityNet. While the technical improvements are well-documented, the precise numerical improvements over Qwen2.5-VL appear not to be fully detailed in publicly available sources as of June 2026.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

26. assistant (2026-06-09T11:27:58.923Z)

I now have comprehensive evidence. Key clinching data points:

- **Off-the-shelf VLM is weak at strict boundaries**: Qwen2.5-VL-7B (the base in the project's plan) gets only $R@0.7=24.3\%$ on Charades-STA out of the box. Fine-tuning/RL (Time-R1) lifts it to 50.1%, but that's still only half of moments correctly localized at $IoU=0.7$.
- **Specialized heads win on boundary precision**: ActionFormer ~66.8% mAP on the harder multi-instance THUMOS14; the field consensus (TimeRefine, "perception decoder," TimeExpert) is that LLMs need a specialized decoder/refinement to get fine boundaries.
- **Multi-sport single model is validated** (RacketVision, MS-TrackNetV3): unified training BOOSTS generalization (+15-19% mAP) but costs slight per-sport pinpoint precision ? exactly the project's tension, with a clean answer.
- **VLM grounding data recipe is concrete** (TVG-R1): ~13K SFT cold-start + ~18K RL, and the public CoT labels were Gemini-2.5-Pro generated ? forbidden here, confirming the no-paid-LLM penalty is real.
- **Conversational QA architecture**: AVAS/event-knowledge-graph + RAG is the emerging best practice for the QA use case, NOT a single end-to-end VLM ? which maps perfectly onto the project's "bookkeeping from the start" requirement.

I have everything needed for a high-confidence recommendation. Let me compose the structured output.

27. user (2026-06-09T11:29:33.124Z)

Structured output provided successfully

28. assistant (2026-06-09T11:29:48.310Z)

I've completed the research and returned my recommendation via StructuredOutput.

Summary of the recommendation: **BOTH, strictly sequenced (Option C)** ? which sharpens the project's existing 3-generation blueprint rather than overturning it.

The decisive evidence: a fine-tuned VLM does **not** reach very-high precision/recall at strict temporal boundaries. The best 2025-26 RL-post-trained video VLM (Time-R1 on Qwen2.5-VL-7B) hits only $R@0.7=50.1\%$ on Charades-STA *fine-tuned*; the same base off-the-shelf is $R@0.7=24.3\%$. Specialized TAL heads (ActionFormer ~66.8% mAP, TBT-Former 68.0% Dec-2025 on harder THUMOS14) stay ahead on boundary precision, and the survey literature is explicit that VLMs "capture coarse regions, not accurate boundaries." Since boundary accuracy is what this product sells, the head is the quality-first extractor NOW; the VLM is the deferred conversational target.

Three course-corrections backed by fresh sources:

1. **Build the per-video event-index/bookkeeping framework now** ? no-regret; the QA loop needs durable per-video state regardless.
2. **Conversational QA should be RAG over the head's event index** (AVAS-style event knowledge graph), not an end-to-end VLM ? the 2025-26 best practice and deterministic for "make a clip of all service faults."
3. **Single sport-agnostic model is feasible and beneficial** ? RacketVision (AAAI'26) shows unified badminton+tennis+TT training improves generalization +15-19% mAP at modest pinpoint cost. The real multisport blocker is clean per-sport trajectory detectors + labeled data + the WASB C3 provenance, not the model architecture.

One concrete guardrail finding worth flagging: the proven VLM grounding recipe (TVG-R1: ~13K SFT + ~18K RL) relies on Gemini-2.5-Pro-generated CoT labels ? forbidden by this project's no-paid-LLM-training rule ? so the VLM path carries a real, binding cost penalty that further justifies deferring it.

Relevant files I grounded against: `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` and `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\research\OWNED\_MODEL\_TRAINING\_STUDY.md`.